

Arnab Kar Sarkar

AI Architecture | Agent Platforms | AI Security & Agent Authorization
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SUMMARY

AI Architect with 16+ years building enterprise platforms and production systems, including three years defining and delivering Okta's company-wide AI architecture. As the first dedicated AI engineer there, I built the foundation the company now runs on: a governed LLM access layer serving ~500M tokens/month across ~7,000 employees, a multi-agent action layer running 15+ production agents for 10+ teams, and a retrieval platform with fine-grained authorization enforced at query time. I set the shared interfaces, reference architectures, and engineering standards those teams build against, and drove adoption hands-on across independent organizations. My depth is in agent systems, AI security and agent authorization, evaluation, and the distributed-systems problems that surface when AI moves from prototype into production.

EXPERIENCE

Architect, AI & Automation — Okta

Nov 2020 – Present

- **Multi-Agent Orchestration Platform.** Led design and production deployment of a LangGraph-based multi-agent platform adopted company-wide as the internal action layer, serving ITSM, customer support, and finance on shared orchestration infrastructure. Architected a skills-based orchestration model where reusable Skills map to tools and any agent can be exposed as a tool to another agent via MCP — an orchestrator routes work across a reasoner, skill executor, and reviewer. Built two-tier routing: utterance embeddings give a deterministic fast path at p50 under 10ms, with LLM routing as fallback for ambiguous queries. Implemented human-in-the-loop interrupt handling and shared state management across the agent graph. Estimated ~100K manual hours saved in recent months from orchestration alone (~325K cumulative since launch in late 2024).
- **Agent Authorization & Execution Boundaries (Auth0 / Okta Token Vault).** Architected and built the enterprise plugin governing how AI agents access enterprise APIs (Salesforce, ServiceNow, Google): credential vaulting, before-tool-call guardrails on credential access, and TOTP-based approval flows for high-risk write operations. Defined the trust boundary between agent autonomy and privileged action, with every action traceable to both the agent and the human it acted on behalf of.
- **Centralized LLM Gateway (Company-Wide Adoption).** Architected and drove company-wide adoption of a centralized LLM gateway (LiteLLM), now the governed access layer for all AI workloads — ~500M tokens/month across ~7,000 employees, with model routing, provider abstraction, and usage governance. Replaced fragmented direct-to-provider calls with a single observable, policy-controlled entry point; drove migration through hands-on pairing, architecture reviews, and internal enablement.
- **RAG-as-a-Service (Internal Platform).** Designed and shipped a company-wide platform letting any team stand up retrieval-augmented LLM applications on shared infrastructure. Built an ACL-aware vector embedding pipeline integrated with Auth0 FGA so fine-grained authorization is enforced at query time, plus a standardized evaluation harness for retrieval and generation quality. Consolidated multiple ad-hoc RAG implementations into one supported platform.
- **No-Code Agent Platform & MCP Tool Layer.** Built a self-service no-code agent platform — ReAct agents with configurable tools and RAG sources, database-backed configuration exposed through a Studio UI — so non-engineers can build production agents without code. Adopted by 10+ teams running 15+ production agents. Stood up a centralized MCP server gateway giving agents grounded, governed access to enterprise tools and team-specific context.
- **Executive Advisory & Architecture Strategy.** Partner directly with Okta's COO, SVPs, and Product Management leadership to define long-term AI strategy and ROI-driven roadmap prioritization. Drive technical due diligence on build-vs-buy decisions for AI capabilities, and resolve cross-team tradeoffs on what becomes shared platform versus where teams retain flexibility.
- **Responsible AI, Governance & Standards.** Established AI guardrails using AWS Bedrock Guardrails for responsible-AI compliance and model interpretability across deployed agents. Codified reference architectures, coding and design standards for AI projects now reused across internal teams. Led organization-wide AI/ML workshops and mentor cross-functional engineers on multi-agent design, RAG architecture, and AI security patterns.
- **Generative AI for Salesforce, ITSM & Enterprise Search.** Architected and deployed enterprise GenAI across Salesforce and ITSM workflows using OpenAI, AWS Bedrock, LangGraph, and Snowflake — improved automation throughput on served workflows by ~90% and built an intelligent case routing agent that improved triage efficiency by ~70%. Shipped enterprise AI search on Coveo and AWS OpenSearch with multi-cloud LLM providers (Anthropic Claude, Google Vertex AI, OpenAI) and ACL-based vector embeddings.

Lead Application Developer — Thumbtack

Apr 2018 – Oct 2020

- Built and deployed ML classification models on AWS for case routing and lead prediction; deployed a Random Forest model for BDR meeting prediction that improved team productivity by ~60%.
- Spearheaded multi-million-record migrations and data pipelines using GCP BigQuery and Apache Spark; developed stream-based APIs for high-volume Salesforce data integration with the data warehouse.
- Implemented CI/CD pipelines and automated testing frameworks for ML models and enterprise applications.

Senior Application Developer — NICE inContact

Dec 2015 – Mar 2018

- Led development of internal ML-based community tools using C# .NET and Angular.
- Built custom SAML integrations and secured enterprise identity management solutions.
- Designed and shipped end-to-end project management applications in Salesforce, increasing team productivity by ~80%.

Earlier Roles

2008 – 2015

- **Cognizant / SelectHealth (2014–2015).** Led HIPAA-compliant Salesforce data migration of 700K+ member records; architected MDM solutions for enforced data quality and member consolidation.
- **Moody's Investors Service (2008–2014).** Salesforce Project Lead managing implementations across the full project lifecycle, leading a 4-person team.

SPEAKING, OPEN SOURCE & PUBLICATIONS

- **OpenClaw Okta Guardian — Oktane (Speaker).** Presented to Okta customers at the company's flagship conference on securing agent execution: trust boundaries, scoped credentials, and approval flows for high-risk agent actions. Open-sourcing shortly.
- **PrivacyFirewall — Local-First DLP Chrome Extension (2024 – Present).** Open-source PII detection running fully in-browser. v2.0 rebuilt with ONNX Runtime Web and Transformer.js for local inference, eliminating server dependency. 225+ GitHub stars. github.com/privacyshield-ai/privacy-firewall
- **Enterprise RAG: What Breaks at Scale — Technical Series (2025 – Present).** Ongoing open series documenting production failure modes in enterprise RAG: chunking strategies, embedding drift, ACL enforcement at retrieval time, and relevance degradation at scale.

TECHNICAL SKILLS

- **AI / ML & GenAI:** LLM application architecture (OpenAI, Anthropic Claude, Google Gemini), multi-agent orchestration (LangGraph, LangSmith), agentic workflows, tool use / function calling, MCP, context engineering, structured outputs, RAG at scale, agent evaluation, prompt engineering, model routing & LLM gateways (LiteLLM), ONNX Runtime Web.
- **Systems & Performance:** Distributed system design, API design, latency and throughput optimization, caching strategies, inference serving and model routing, cost-per-call optimization, high-volume data pipelines.
- **AI Security & Governance:** Agent authorization and execution boundaries, scoped credentials and credential vaulting, prompt injection mitigation, ACL-aware retrieval, agent action approval flows, AWS Bedrock Guardrails, PII detection, auditability of agent actions.
- **Cloud & Infrastructure:** AWS (Bedrock, Lambda, API Gateway, ECS, S3, OpenSearch), GCP (BigQuery, Vertex AI), Kubernetes, Docker.
- **Languages & Platforms:** Python, Go, Node.js, TypeScript, Java, React; Salesforce (Apex, LWC, FFLIB, Platform Events), ServiceNow, Snowflake, Auth0 / Auth0 FGA.
- **MLOps:** A/B testing for AI, canary releases, evaluation harnesses for non-deterministic systems, CI/CD for ML.

EDUCATION

M.S. Computer Science — University of Illinois Urbana-Champaign
B.S. Computer Science — West Bengal University of Technology

GPA 3.91 | Dec 2022
Aug 2008